STAT 361 Assignment 1 Question 2

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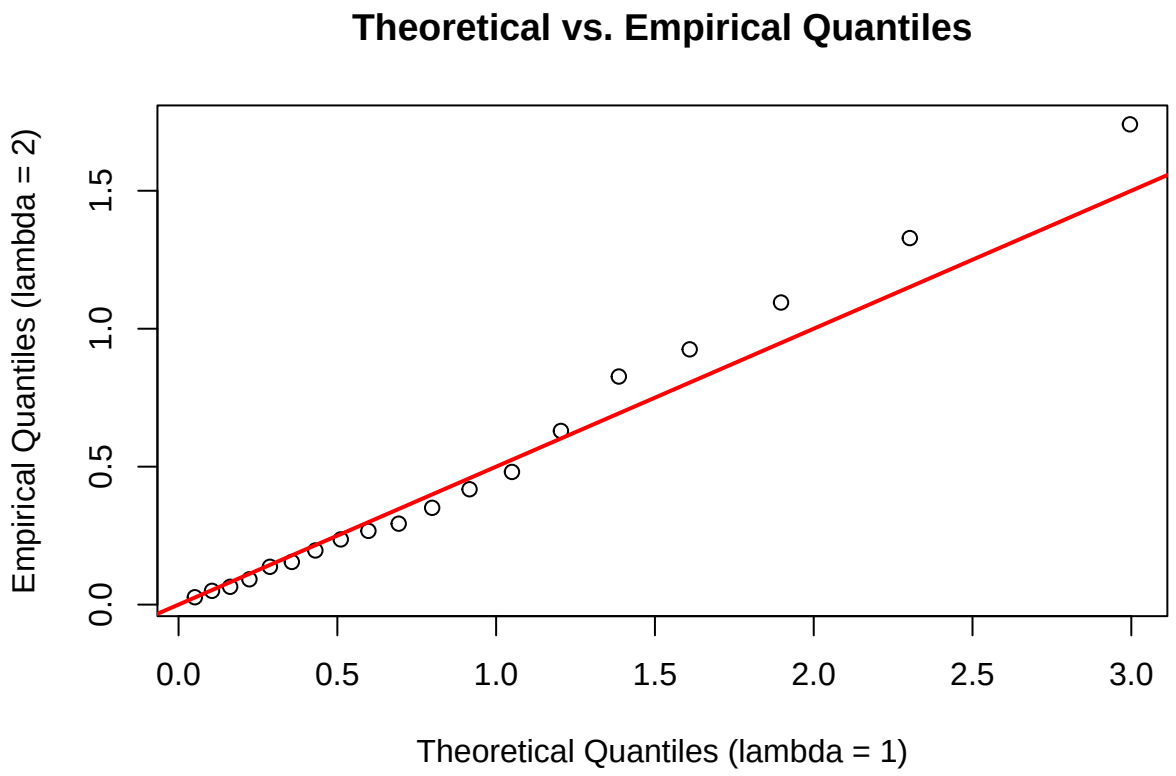
a) Random number generation and empirical quantile finding

```
randnums <- rexp(200, rate = 2)
empqtiles <- quantile(randnums, probs = seq(0.05, 0.95, 0.05))
```

b) Theoretical quantile finding and quantile plots

```
thqtiles = qexp(p = seq(0.05, 0.95, 0.05), rate = 1)
plot(thqtiles, empqtiles, xlab = "Theoretical Quantiles (lambda = 1)", ylab = "Empirical Quantiles (lam

abline(a = 0, b = 0.5, col = "red", lwd = 2) # Rate of 0.5 since exp rates are 1/2 ratio
```



c) Expected line

Shown in part b) using `abline()`. Expected a slope of 0.5 as the two distributions are the same, just with a different lambda which is directly proportional in terms of scaling differences amongst exponential functions. The ratio is simply 1/2 or 0.5, hence the slope being 0.5.